

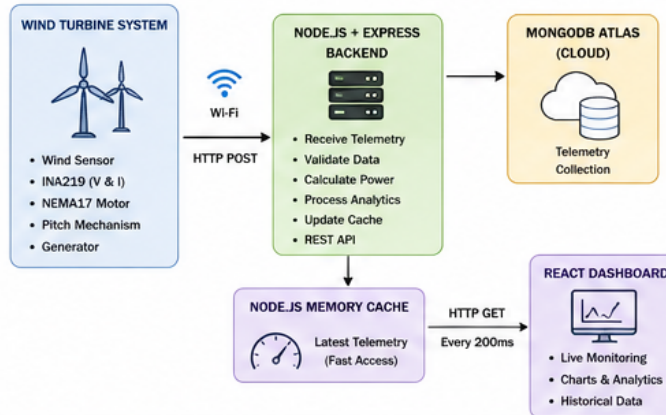
Wind Turbine IoT Monitoring System — Architecture & Data Flow

WIND TURBINE IoT MONITORING SYSTEM

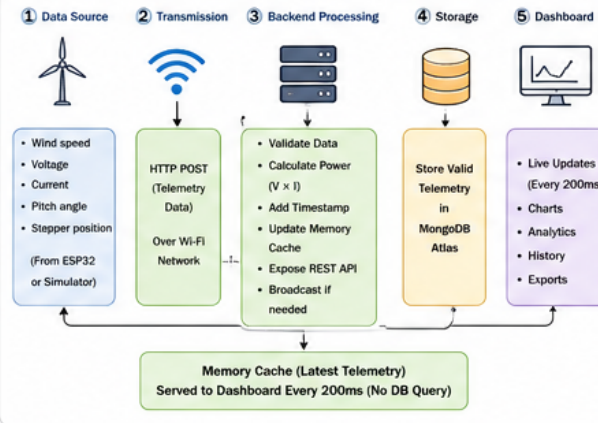
SYSTEM ARCHITECTURE & DATA FLOW DIAGRAMS



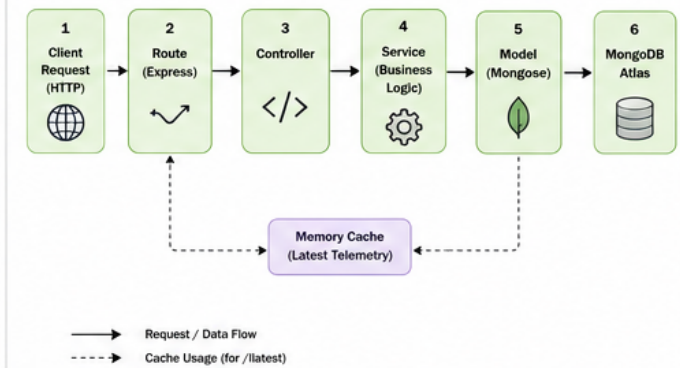
1. OVERALL SYSTEM ARCHITECTURE



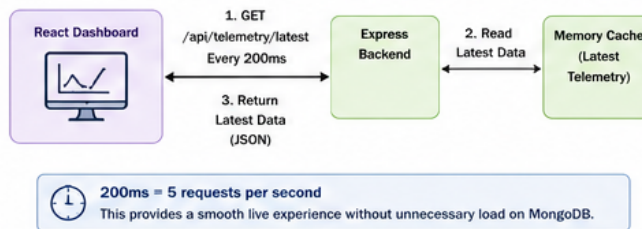
2. COMPLETE DATA FLOW DIAGRAM



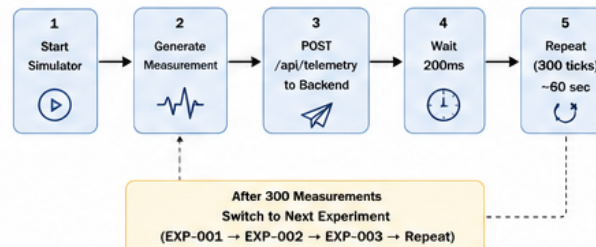
3. BACKEND REQUEST FLOW



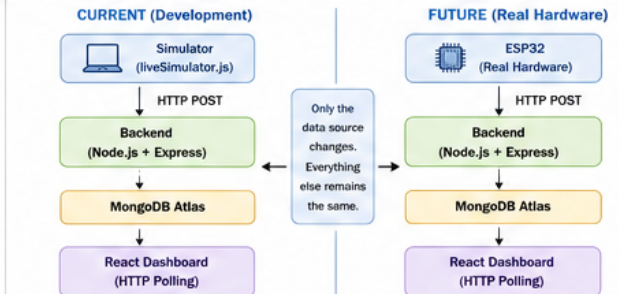
4. DASHBOARD LIVE-UPDATE FLOW (200ms POLLING)



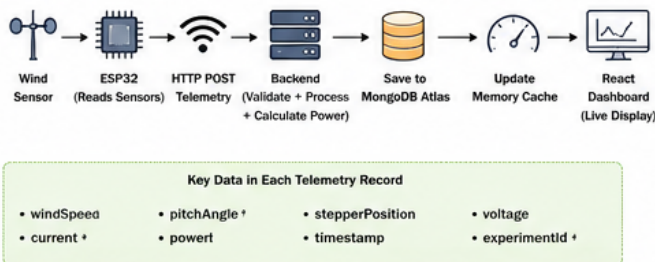
5. SIMULATOR FLOW (DEVELOPMENT MODE)



6. CURRENT VS FUTURE ARCHITECTURE



7. DATA FLOW – FROM SENSOR TO DASHBOARD



8. COMMUNICATION OVERVIEW

Communication	Protocol	Direction	Purpose	Used By	Frequency
Telemetry Transmission	HTTP (POST)	ESP32 / Simulator → Backend	Send measurement data to backend	Hardware / Simulator	Every 200ms (configurable)
Live Data Retrieval	HTTP (GET)	React → Backend	Get latest telemetry from memory cache	Dashboard	Every 200ms (5 req/sec)
Dashboard Other APIs	HTTP (GET)	React → Backend	History, analytics, summary, etc.	Dashboard	On demand
Database Storage	MongoDB Driver	Backend → MongoDB	Persist valid telemetry	Backend	Per received measurement

9. PITCH ANGLE & EXPERIMENTS

